

Companion Document: update_F.R

q(F) — Variational Update for Genomics Loading Matrix

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Table of contents

1	Overview	2
2	Mathematical Background	2
2.1	Model Structure	2
2.2	EBNM Sufficient Statistics for $f_{\cdot}$	2
3	The Tau Cancellation Property	3
3.1	Tau Cancels in $x_{\cdot j}$ (observation)	3
3.2	Tau Does NOT Cancel in $s_{\cdot j}$ (noise level)	3
3.3	Consequence	4
4	API Reference	4
4.1	update_F_k(Tau, EL_k, EL2_k, R_k, prior_family, A_floor)	4
4.2	update_F_all(Y, EL, EL2, EF, EF2, Tau, prior_family, A_floor)	5
5	Test Suite	6
5.1	T1: Mathematical Identity Checks (6 tests)	6
5.2	T2: WLS / Non-Informative Limit (2 tests)	6
5.3	T3: Known Signal Recovery K=1 (3 tests)	7
5.4	T4: Multi-Factor K=5 (3 tests)	7
5.5	T5: Null Factor Shrinkage (2 tests)	7
5.6	T6: Tau Differential Shrinkage (3 tests)	7
5.7	T7: Numerical Stability (4 tests)	7
5.8	T8: $R_{\cdot k}$ and Gauss-Seidel (2 tests)	8
5.9	T9: V2.R Consistency (1 test)	8
6	Demos	8
6.1	Demo 1: Tau Cancellation Property	8
6.2	Demo 2: Signal Recovery K=1	9
6.3	Demo 3: Tau-Dependent Shrinkage	9
6.4	Demo 4: Sparsity Recovery	9
6.5	Demo 5: Multi-Factor K=5	10

7 Computational Notes	10
7.1 Scalar Broadcast	10
7.2 Shared Dependency: compute_R_k	10
7.3 Memory Layout	10
8 Related Files	10

1 Overview

update_F.R implements the CAVI variational update for the factor loading matrix **F** (shape $p \times K$) in the joint genomics + survival model. **F** appears **only** in the genomics likelihood and has no role in the Cox proportional hazards survival term — this structural fact drives the module’s key mathematical simplification.

The update is a **vector EBNM** (Empirical Bayes Normal Means) problem solved independently for each factor column $f_{\{.k\}}$. Within each column, the p feature-wise posterior means and variances are computed in closed form, then passed to an EBNM solver that applies the chosen prior (default: point-normal) and returns posterior moments.

Depends on: compute_R_k() from code/update_L.R — source that file before this one.

2 Mathematical Background

2.1 Model Structure

The genomics observations follow:

$$Y = L F' \quad (n \times p \text{ matrix, } L \text{ is } n \times K, F \text{ is } p \times K)$$

with feature-wise precision $\text{Tau}[j]$ (heteroscedastic noise). The residual for factor k at feature j is:

$$R_{k[i,j]} = Y[i,j] - \sum_{\{k' \neq k\}} EL[i,k'] * EF[j,k']$$

This is computed by compute_R_k() and excludes the current factor k (Gauss-Seidel schedule).

2.2 EBNM Sufficient Statistics for $f_{\{.k\}}$

For each feature j , the ELBO contribution from factor k leads to the following normal sufficient statistics:

Quantity	Formula	Shape	Notes
$A_F[j]$	$\text{Tau}[j] * \text{sum}(\text{EL2_k})$	p-vector	$\text{sum}(\text{EL2_k})$ is a scalar broadcast across all j
$B_F[j]$	$\text{Tau}[j] * (\text{t}(\text{R_k}) \% \% \text{EL_k})[j]$	p-vector	Inner product of residual column with L posterior
$x_F[j]$	$B_F[j] / A_F[j]$	p-vector	EBNM observation ("sufficient statistic")
$s_F[j]$	$1 / \text{sqrt}(A_F[j])$	p-vector	EBNM noise standard deviation

$\text{sum}(\text{EL2_k})$ is the sum of the second moments of the k-th column of L across all n samples:

$$\text{sum_EL2_k} = \text{sum_i EL2}[i,k] \quad (\text{scalar, includes } \text{Var_q}(l_{\{ik\}}) + \text{EL}[i,k]^2)$$

3 The Tau Cancellation Property

This is the **central mathematical insight** of the module.

3.1 Tau Cancels in x_j (observation)

$$\begin{aligned} x_j &= B_F[j] / A_F[j] \\ &= (\text{Tau}[j] * (\text{t}(\text{R_k}) \% \% \text{EL_k})[j]) / (\text{Tau}[j] * \text{sum_EL2_k}) \\ &= (\text{t}(\text{R_k}) \% \% \text{EL_k})[j] / \text{sum_EL2_k} \end{aligned}$$

The $\text{Tau}[j]$ factors in numerator and denominator cancel exactly. The EBNM observation x_j is **identical regardless of feature precision**.

3.2 Tau Does NOT Cancel in s_j (noise level)

$$s_j = 1 / \text{sqrt}(A_F[j]) = 1 / \text{sqrt}(\text{Tau}[j] * \text{sum_EL2_k})$$

Here $\text{Tau}[j]$ remains under the square root. A feature with higher precision (larger $\text{Tau}[j]$) gets a **smaller** noise level s_j , which means the prior applies less shrinkage and the posterior mean stays closer to x_j .

3.3 Consequence

Features with the same residual signal x_j are differentially shrunk toward zero depending on their noise precision. Clean features (high τ) are shrunk less; noisy features (low τ) are shrunk more. This is the heteroscedastic shrinkage mechanism for F .

τ	x_j	s_j	Shrinkage
1	same	large	strong
1000	same	small ($\sim 1/\sqrt{1000}$)	smaller) weak

Demo 1 illustrates this with $\max |x_j - \text{diff}| = 0$ and $s_j \propto \sqrt{1000}$.

4 API Reference

4.1 `update_F_k(Tau, EL_k, EL2_k, R_k, prior_family, A_floor)`

Computes the variational posterior for a single factor column $f_{\cdot k}$.

Arguments:

Parameter	Type	Description
τ	numeric vector, length p	Feature-wise noise precisions
EL_k	numeric vector, length n	Posterior means of L column k : $E_q[l_{\cdot k}]$
$EL2_k$	numeric vector, length n	Posterior second moments of L column k : $E_q[l_{\cdot k}^2]$
R_k	numeric matrix, $n \times p$	Residual matrix excluding factor k (from <code>compute_R_k</code>)
<code>prior_family</code>	character	EBNM prior family; default "point_normal"
A_{floor}	numeric	Minimum value for A_F to avoid division by zero; default $1e-10$

Returns: Named list

Field	Type	Description
<code>mean</code>	numeric vector, length p	Posterior mean $E_q[f_{\cdot k}]$
<code>second</code>	numeric vector, length p	Posterior second moment $E_q[f_{\cdot k}^2]$

Field	Type	Description
sd	numeric vector, length p	Posterior standard deviation
A	numeric vector, length p	EBNM precision A_F (after floor)
B	numeric vector, length p	EBNM numerator B_F
x	numeric vector, length p	EBNM sufficient statistic $x_F = B/A$
s	numeric vector, length p	EBNM noise $s_F = 1/\sqrt{A}$
sum_EL2_k	numeric scalar	sum(EL2_k), shared across all features
ebnm_result	list	Raw output from ebnm() call

Key implementation details:

- $A_F = \text{pmax}(\text{Tau} * \text{sum}(\text{EL2_k}), A_{\text{floor}})$ — the floor prevents NaN when Tau is zero
- $B_F = \text{Tau} * (\text{t}(\text{R_k}) \%*\% \text{EL_k})$ — matrix-vector product gives p-vector
- Second moment satisfies: $\text{second}[j] = \text{sd}[j]^2 + \text{mean}[j]^2$

4.2 update_F_all(Y, EL, EL2, EF, EF2, Tau, prior_family, A_floor)

Gauss-Seidel loop over all K factor columns.

Arguments:

Parameter	Type	Description
Y	numeric matrix, $n \times p$	Observed genomics data
EL	numeric matrix, $n \times K$	Posterior means of L (fully updated from L step)
EL2	numeric matrix, $n \times K$	Posterior second moments of L
EF	numeric matrix, $p \times K$	Current posterior means of F (updated in place)
EF2	numeric matrix, $p \times K$	Current posterior second moments of F
Tau	numeric vector, length p	Feature-wise precisions
prior_family	character	EBNM prior family; default "point_normal"
A_floor	numeric	Floor for A; default 1e-10

Returns: Named list

Field	Type	Description
EF	numeric matrix, $p \times K$	Updated posterior means of F
EF2	numeric matrix, $p \times K$	Updated posterior second moments of F
details	list of length K	Per-factor output from <code>update_F_k</code>

Important Gauss-Seidel note:

`update_F_all` passes the **fully updated** EL (not a stale copy) because the L step completes before the F step in the CAVI schedule. Within the F loop, `EF_curr` is updated column-by-column so that each `compute_R_k` call sees the most recent estimates of all other F columns. This is the Gauss-Seidel (sequential) schedule, as opposed to a Jacobi (parallel) schedule.

```
for k in 1:K:
  R_k <- compute_R_k(Y, EL, EF_curr, k) # excludes column k
  res <- update_F_k(Tau, EL[,k], EL2[,k], R_k, ...)
  EF_curr[,k] <- res$mean
  EF2_curr[,k] <- res$second
```

5 Test Suite

File: `tests/test_update_F.R` Total: **26 tests across 9 groups** — all passing.

5.1 T1: Mathematical Identity Checks (6 tests)

Test	What is verified
T1.1	$A_F[j] = \text{Tau}[j] * \text{sum}(EL2_k)$ exactly
T1.2	$B_F[j] = \text{Tau}[j] * (\text{t}(R_k) \%*\% EL_k)[j]$ exactly
T1.3	x is a p -vector (length p)
T1.4	Tau cancels in x_j : $\text{Tau}=1$ vs $\text{Tau}=1000$ produce identical x_j
T1.5	Tau does NOT cancel in s_j : s_j ratio is significantly different
T1.6	Second moment identity: $\text{second}[j] = \text{sd}[j]^2 + \text{mean}[j]^2$

5.2 T2: WLS / Non-Informative Limit (2 tests)

Test	What is verified
T2.1	When $EL2_k = EL_k^2$ (zero posterior variance in L), x_j equals the OLS estimator
T2.2	Large τ drives posterior mean close to the observation x_j

5.3 T3: Known Signal Recovery $K=1$ (3 tests)

Test	What is verified
T3.1	Dense factor: correlation between recovered and true f exceeds 0.7
T3.2	Sparse factor (50/500 nonzero): active features have larger posterior mean than inactive
T3.3	Higher $\tau \rightarrow$ better recovery (monotone precision effect)

5.4 T4: Multi-Factor $K=5$ (3 tests)

Test	What is verified
T4.1	Output matrices EF , $EF2$ have shape $p \times K$
T4.2	All second moments $\geq$ squared means ($EF2 \geq EF^2$) component-wise
T4.3	All outputs are finite (no NaN or Inf)

5.5 T5: Null Factor Shrinkage (2 tests)

Test	What is verified
T5.1	Pure-noise R_k produces factor near zero (point-normal shrinks to zero)
T5.2	Zero $EL_k \rightarrow A$ hits floor, $B=0 \rightarrow EF$ near zero

5.6 T6: Tau Differential Shrinkage (3 tests)

Test	What is verified
T6.1	High- τ features are shrunk less than low- τ features (same x_j)
T6.2	Larger $EL2 \rightarrow$ larger $A \rightarrow$ smaller $s \rightarrow$ less shrinkage for high-signal factors
T6.3	Uniform $\tau \rightarrow$ all s_j equal (homoscedastic limit, one shared noise level)

5.7 T7: Numerical Stability (4 tests)

Test	What is verified
T7.1	All-zero Tau $\rightarrow$ A hits floor, no NaN or Inf in output
T7.2	Extreme Tau=1e10 $\rightarrow$ no overflow in A, B, x, s
T7.3	Edge case p=1, n=1 $\rightarrow$ correct scalar outputs
T7.4	Custom A_floor value is respected ($A \geq A_floor$)

5.8 T8: R_k and Gauss-Seidel (2 tests)

Test	What is verified
T8.1	R_k correctly uses EL_k (not another factor's column)
T8.2	update_F_all produces finite results with valid second moments across all K

5.9 T9: V2.R Consistency (1 test)

Test	What is verified
T9.1	update_F_k output matches the corresponding lines in Supervised_Bayesian_MF_V2.R (lines 334–340) exactly

6 Demos

File: demos/demo_update_F.R Total: **5 scenarios** — narrative output, no PASS/FAIL.

6.1 Demo 1: Tau Cancellation Property

Purpose: Demonstrate the core mathematical property that τ_j cancels in x_j but not in s_j .

Setup: Same Y, L, R_k. Run update_F_k twice — once with $\text{Tau} = \text{rep}(1, p)$, once with $\text{Tau} = \text{rep}(1000, p)$.

Expected output: $-\max_j |x_j(\text{Tau}=1) - x_j(\text{Tau}=1000)| \approx 0$ (machine precision) - $s_j(\text{Tau}=1) / s_j(\text{Tau}=1000) \approx \sqrt{1000} \approx 31.6$ for all j

This illustrates that the EBNM “data” is precision-free, while the EBNM “noise level” encodes the precision.

6.2 Demo 2: Signal Recovery $K=1$

Purpose: Confirm that the update recovers a sparse true factor from noisy observations.

Setup: $n=150$, $p=200$, $K=1$. True f_{true} is sparse (30/200 nonzero entries, drawn from $N(0,1)$). $Y = L \%* \% t(f_{\text{true}}) + \text{noise}$.

Reported metrics: - Pearson correlation between $EF[:,1]$ and f_{true} - Mean posterior mean for active vs inactive features (active should be larger)

6.3 Demo 3: Tau-Dependent Shrinkage

Purpose: Show that heteroscedastic precision differentially regularizes features.

Setup: $p=200$ features split into two groups: first 100 with $\tau=10$ (clean), last 100 with $\tau=0.5$ (noisy). Same true signal in both groups.

Reported metrics: - Mean absolute error (MAE) for clean vs noisy features - Posterior SDs for clean vs noisy features - Demonstrates that clean features (high τ) stay closer to truth

6.4 Demo 4: Sparsity Recovery

Purpose: Evaluate the point-normal prior's ability to identify which features belong to a sparse factor.

Setup: $n=100$, $p=250$, true f is sparse (40/250 nonzero). Threshold posterior means at 0.1 to classify active/inactive.

Reported metrics (confusion matrix):

	Predicted Active	Predicted Inactive
True Active	TP	FN
True Inactive	FP	TN

Also reports precision = $TP/(TP+FP)$ and recall = $TP/(TP+FN)$.

6.5 Demo 5: Multi-Factor K=5

Purpose: Validate `update_F_all` on a structured 5-factor problem.

Setup: $n=200$, $p=300$, $K=5$. Each factor has a distinct sparse structure; F is initialized at zero.

Reported metrics: - Column-wise correlations: $\text{cor}(EF[:,k], F_{\text{true}}[:,k])$ for each k - Number of active features per factor (posterior mean > 0.1) - Confirms Gauss-Seidel within-loop updates do not break structure

7 Computational Notes

7.1 Scalar Broadcast

`sum(EL2_k)` is computed once (a single scalar) and broadcast across all p features via element-wise multiplication with Tau . This avoids a $p \times n$ matrix multiply for A_F .

7.2 Shared Dependency: `compute_R_k`

`compute_R_k` is defined in `code/update_L.R`. Always source `update_L.R` before `update_F.R`. The residual matrix R_k has shape $n \times p$ and represents the data with all other factors' contributions removed. Because L is fully updated before the F step, the R_k computation here uses the most recent L posterior.

7.3 Memory Layout

Matrix	Shape	Update order
EL	$n \times K$	Fixed (updated in L step)
EL2	$n \times K$	Fixed (updated in L step)
EF	$p \times K$	Updated column by column (Gauss-Seidel)
EF2	$p \times K$	Updated column by column
R_k	$n \times p$	Recomputed fresh for each k

For large p , R_k ($n \times p$) dominates memory. It is not cached between factor updates.

8 Related Files

File	Relationship
code/update_L.R	Defines compute_R_k() required by this module
code/update_tau.R	Provides Tau used as input here
code/update_beta.R	Beta update (survival); F does not appear there
code/Supervised_Bayesian_MF_Main.R	Main Algorithm; lines 334–340 are the reference for T9.1
tests/test_update_F.R	26 tests, all passing
demos/demo_update_F.R	5 narrative demos
derivations/qF/qF_update_derivation.qF	Full derivation including tau cancellation proof
code/SupervisedMF_Context.m4	AI/code quick-reference for the full R implementation